

THE ESSENTIALS OF ALGEBRAIC NUMBER THEORY

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1. ARITHMETICS IN $\mathbb{Z}$

In this document we give proofs of crucial elementary results in number theory. The first theorem that we shall cover is a lemma due to Bézout:

Proposition 1 (Bézout Lemma). *Let a, b be non-zero integers. There exist integers u and v such that $\gcd(a, b) = ua + vb$.¹*

PROOF. Consider the set $\mathcal{E} := \{xa + yb > 0 : x, y \in \mathbb{Z}\}$. We first make the observation that this set is non-empty, for taking $x = a$ and $y = b$ we obtain an element of $\mathcal{E}$, $a^2 + b^2 > 0$, proving that $\mathcal{E}$ is non-void. Now, let $d \in \mathcal{E}$ be the least element. We claim that $d \mid e$ for all $e \in \mathcal{E}$. Indeed, let us write

$$e = qd + r, \quad 0 \leq r < d \quad (1)$$

Let us assume, by way of contradiction, that $d \nmid e$. Then, $r > 0$ and we may write

$$0 < r = e - qd = (\mu a + \nu b) - q(ua + vb) = (\mu - qu)a + (\nu - qv)b < d$$

is an element in $\mathcal{E}$ less than d : a contradiction. Ergo, $d \mid e$ and in particular $d \mid a, b$. We are now prepared to show $d \leq \gcd(a, b)$. Certainly, let c be a common divisor of a, b . Then, $a = q_a c$ and $b = q_b c$ and we may write by definition of $d \in \mathcal{E}$:

$$d = ua + vb = u(q_a c) + v(q_b c) = c(uq_a + vq_b) \text{ implying } c \mid d$$

We have shown that $d \mid a, b$ and any common divisor of a, b divides d : thence we find that $d = \gcd(a, b)$ by definition. The proof is complete. ○

A more interesting theorem is that of Fermat:

Theorem 2 (Fermat). *Let $p \in \mathbb{Z}$ be a prime. Then, $x^{p-1} \equiv 1 \pmod{p}$ for all non-zero $x \in \mathbb{Z}/p\mathbb{Z}$.*

¹This proof does not tell one how to compute the coefficients $u, v \in \mathbb{Z}$. These integers may be found by following the Euclidean Algorithm and reverse-substituting the remainders.

PROOF. Consider the equivalence classes of $\mathbb{Z}/p\mathbb{Z}$. Since p is prime, this ring is a field and hence there exists an inverse for any $x \neq 0$. Define a mapping

$$\phi : (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times, \quad [a] \mapsto [a \cdot x] \quad (2)$$

where $[\cdot]$ denotes an equivalence class modulo p . We claim this mapping is a bijection. Certainly, since x is invertible the auxiliary map

$$\psi : (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times, \quad [a] \mapsto [a \cdot x^{-1}]$$

is well defined. Moreover, it is easy to verify that $\phi \circ \psi = \psi \circ \phi = \text{id}$ and hence we find that ϕ (and ψ) are bijections. Loosely speaking, all that ϕ does is “shuffle” the equivalence classes of $(\mathbb{Z}/p\mathbb{Z})^\times$. Thus,

$$[1] \cdot [2] \cdot [3] \cdots [p-1] = [x] \cdot [2x] \cdot [3x] \cdots [x(p-1)] = [x^{p-1}][(p-1)!]$$

Or,

$$x^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$$

since $(p-1) \not\equiv 0$ in $\mathbb{Z}/p\mathbb{Z}$ we may divide this out, to find

$$x^{p-1} \equiv 1 \pmod{p}$$

○

Theorem 3 (Wilson). *Let $p \in \mathbb{Z}$. Then, p is prime if and only if $(p-1)! \equiv -1 \pmod{p}$.*

PROOF. First we assume p is a prime. If $p = 2, 3$ we may verify this mechanically, and thus we may assume without harm to the proof that $p \geq 5$. Note that $(-1)^2 \equiv 1^2 \equiv 1$ in $\mathbb{Z}/p\mathbb{Z}$. We claim that these are the only elements with this property. Certainly, if $x^2 \equiv 1 \pmod{p}$ then $p \mid (x-1)(x+1)$. Since $\mathbb{Z}/p\mathbb{Z}$ is an integral domain, either $x \equiv -1$ or $x \equiv 1$. In light of this, if we consider the list

$$S := \{2, 3, \dots, p-2\}$$

containing *evenly* many elements, we note that each element $s \in S$ has a unique inverse appearing in S . Thus,

$$2 \cdot 3 \cdots p-2 \equiv (p-2)! \equiv 1 \pmod{p} \quad (3)$$

If we multiply (3) by $p-1$ we conclude the proof of one direction.

Conversely, suppose the criterion but assume that p is composite. Namely, $p = ab$ where these integers are such that $1, a, b < p$. Then, we find that $p \mid ab \mid (p-1)!$. Consequently,

$$(p-1)! \equiv 0 \pmod{p}$$

which is absurd. This concludes the proof of the theorem.

○

Theorem 4 (Chinese Remainder Theorem). *Let $\mathbb{Z} \ni n, m$ be co-prime. Then, there exists an integer x solving the congruences*

$$x \equiv a \pmod{n} \quad (4)$$

$$x \equiv b \pmod{m} \quad (5)$$

where $a, b \in \mathbb{Z}$ are given. Moreover, this x is unique up to reduction modulo p .²

²This result may be easily modified to show that $\mathbb{Z}/nm\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.

PROOF. We shall first prove that an integer solution exists. Now, $\gcd(n, m) = 1$ and so n has an inverse n_m in $\mathbb{Z}/m\mathbb{Z}$ and likewise m has a multiplicative inverse m_n in $\mathbb{Z}/n\mathbb{Z}$. Now consider the integer defined below:

$$x := a(m)(m_n) + b(n)(n_m) \quad (6)$$

We claim that this solves the congruences. Certainly, reduction modulo n leaves us with $x \equiv am m_n \equiv a \cdot 1 \equiv a$. Similarly, we find $x \equiv b$ in $\mathbb{Z}/m\mathbb{Z}$ and this concludes the proof of existence.

Now, suppose x, y are two integers solving the equations above. We shall show they belong to the same equivalence class. Clearly, we have

$$x - y \equiv 0 \pmod{n} \text{ implying } n \mid (x - y) \quad (7)$$

$$x - y \equiv 0 \pmod{m} \text{ implying } m \mid (x - y) \quad (8)$$

Now, using that $\gcd(n, m) = 1$ we deduce from the above that $nm \mid x - y$, whence $x - y \equiv 0$ in $\mathbb{Z}/nm\mathbb{Z}$ and $x \equiv y$. This concludes the proof. ○

2. PRIMITIVE ROOTS

Here we give the essentials and bare bones of primitive roots theory. Recall that $(\mathbb{Z}/n\mathbb{Z})^\times$ denotes the set of all elements mod n that are co-prime to n . That is, the set $\{1 \leq x \leq n : \gcd(x, n) = 1\}$. This forms a group under multiplication, as can be easily shown.

We may define the a mapping $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ called the **Euler Totient function** defined by

$$\varphi : \mathbb{N} \rightarrow \mathbb{N}, \quad n \mapsto \# (\mathbb{Z}/n\mathbb{Z})^\times \quad (9)$$

In the above we use the “ $\#$ ” to indicate the *cardinality of a set*³. A primitive root modulo n is an element of order $\varphi(n)$. Hence, if $(\mathbb{Z}/n\mathbb{Z})^\times$ has a primitive root q then $(\mathbb{Z}/n\mathbb{Z})^\times = \langle q \rangle$. An important fact is that $(\mathbb{Z}/p\mathbb{Z})^\times$ has a primitive root whenever p is a prime. We do require the following lemma:

Lemma 5. *Let G be a finite group and $a, b \in G$ with orders n, m respectively. Then, $ab \in G$ has order nm .*

PROOF. Let $\ell = o(ab)$. Note that $\ell \leq nm$. Now, we also see from $(ab)^{nm} = 0$ that $\ell \mid nm$. If we write $\ell = rs \mid nm$ with $r \mid n$ and $s \mid m$. Then, note that

$$e = (ab)^\ell = a^{rs}b^{rs} \text{ implying } e = [a^{rs}b^{rs}]^{\frac{n}{r}} = a^{ns}b^{ns} = b^{ns}$$

so, $m \mid ns$ whence $m \mid s$ since $\gcd(n, m) = 1$. That is, $m = s$. Similarly, we find that $n = r$ and that $\ell = nm$ as was asserted. ○

Lemma 6. *Let f be a polynomial of degree $\deg(f) = n$ over a field $\mathbb{F}$. Then, f has at most $\deg(f)$ roots in $\mathbb{F}$.*

PROOF. We argue by induction. The case $n = 1$ is obvious (f is a linear polynomial). Now assume that this is true if $\deg(g) = n - 1$ and fix a polynomial $f \in \mathbb{F}[x]$ of degree $\deg(f) = n$. We shall show that this again holds (with respect to n).

³Since we are considering finite rings and groups, cardinality is equivalent to the number of representatives.

If f has no-roots we are done. Else, let x_* be a root of the polynomial $f(x)$ and write

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + x_0$$

where we assume without harm that the leading coefficient is 1 ($\mathbb{F}[x]$ is an integral domain). Then, since $f(x_*) = 0$ we may write

$$f(x) = (x - x_*)g(x), \quad g \in \mathbb{F}[x] \text{ with } \deg(g) = n - 1$$

Now, if $\tilde{x}$ is a root of f not equal to x_* then since $\mathbb{F}$ is an integral domain, we must have that $\tilde{x}$ is a root of g . Since g has degree $n - 1$, our induction hypothesis guarantees that g has at most $n - 1$ roots. This concludes the proof. ○

Theorem 7. *Let $p \in \mathbb{N}$ be a prime and $d \mid (p - 1)$. Then $f \in (\mathbb{Z}/p\mathbb{Z})[x]$ given by $f(x) = x^d - 1$ has **exactly** d roots in $\mathbb{Z}/p\mathbb{Z}$.*

PROOF. Set $e = \frac{p-1}{d} \in \mathbb{N}$. Then, we may obviously write

$$\begin{aligned} x^{p-1} - 1 &= x^{de} - 1 = (x^d - 1)(x^{d(e-1)} + x^{d(e-2)} + \cdots + 1) \\ &= (x^d - 1)g(x) \end{aligned}$$

for a polynomial g over this same division ring of degree $\deg(g) = d(e - 1) = de - d = p - 1 - d$. By Lemma 6 we find that g has at most $p - 1 - d$ roots. However, by Fermat's little theorem we have that $x^{p-1} - 1$ has exactly $p - 1$ roots. Now, by this we see that $f(x)$ has *at-least* $p - 1 - (p - 1 - d) = d$ roots. But, again applying Lemma 6 we complete the proof. ○

Theorem 8. *Let p be a prime integer. There exists a primitive root in $(\mathbb{Z}/p\mathbb{Z})^\times$. In particular, the group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic.*

PROOF. If $p = 2$ then $(\mathbb{Z}/2\mathbb{Z})^\times = \{1\} = \langle 1 \rangle$ and so we are done. Suppose without harm that $p \geq 3$. That is, we may find primes q_j such that

$$p - 1 = \prod_{j=1}^n q_j^{e_j} \tag{10}$$

Now, $q_j^{e_j} \mid p - 1$ for all $j = 1, \dots, n$. Therefore, by Theorem 7 there are exactly $q_j^{e_j}$ elements modulo p such that $x^{q_j^{e_j}} \equiv 1$ and $q_j^{e_j-1}$ elements such that $x^{q_j^{e_j-1}} \equiv 1$. Putting this together, we may pickup exactly $q_j^{e_j-1}(q_j - 1)$ elements such that $x^{q_j^{e_j}} \equiv 1$ whilst $x^{q_j^{e_j-1}} \not\equiv 1$.

Now we claim that any such x has order $o(x) = q_j^{e_j}$. Otherwise, if $\ell = o(x)$ then $\ell \mid q_j^{e_j}$ so that ℓ is an integer power of q_j . That is, we may find $f < e_j$ such that $x^{q_j^f} \equiv 1$. But then, for suitable α we find that

$$x^{q_j^{e_j-1}} = x^{q_j^f + \alpha} = x^{q_j^f q_j^\alpha} \equiv 1$$

which is absurd. Hence, x is an element of order $q_j^{e_j}$. We may do so for all such j , and taking the product over such x and successively applying Lemma 5 we complete the proof, since $\varphi(p) = p - 1$. This theorem is now proven. ○

Lemma 9. *Let G be a finite group of order n and suppose that $g \in G$ satisfies $G = \langle g \rangle$. Then, g^m has order $\frac{n}{\gcd(n,m)}$ for all $m \in \mathbb{N}$.*

PROOF. First of all,

$$(g^m)^{\frac{n}{\gcd(n,m)}} = g^{\text{lcm}(n,m)} = 1$$

and thus $\ell = o(g^m) \leq \frac{n}{\gcd(n,m)}$. Now, if $\ell = o(g^m)$ then $g^{m\ell} = e$ which implies that $n \mid m\ell$. Since $m\ell \geq \gcd(n,m)$ and $n \geq \gcd(n,m)$ we pick up

$$\gcd(n,m) \leq n \leq m\ell$$

we obtain upon division that $\frac{n}{\gcd(n,m)} \leq \ell$ which completes this short proof. ○

Proposition 10. *Suppose n is such that there exists a primitive root in $(\mathbb{Z}/n\mathbb{Z})^\times$. Then there are exactly $\varphi(\varphi(n))$ primitive roots in $(\mathbb{Z}/n\mathbb{Z})^\times$.*

PROOF. Let x be a primitive root of $(\mathbb{Z}/n\mathbb{Z})^\times$, i.e. $o(x) = \varphi(n)$. Since any element in $(\mathbb{Z}/n\mathbb{Z})^\times$ is x^m for some m , we may only count m such that $o(x^m) = \frac{\varphi(n)}{\gcd(\varphi(n),m)} = n$. That is, we count m such that $\gcd(\varphi(n),m) = 1$. But the number of such elements is precisely the number of elements in $(\mathbb{Z}/\varphi(n)\mathbb{Z})^\times$. Hence, there are $\varphi(\varphi(n))$ primitive roots. ○

Note that the above proof in fact tells us how to compute all primitive roots from a single one. This is a useful fact, even in practice and one should remember this fact (exams).

3. QUADRATIC RECIPROCITY

We briefly consider linear congruences modulo some prime p . If an equation

$$ax \equiv b \pmod{n}, \quad a, b \neq 0 \text{ and } \gcd(a, n) = 1$$

has a solution then it may found by computing a^{-1} and taking $x = a^{-1}b$. This inverse a^{-1} may be computed via Proposition 1 since $\gcd(a, n)$ are co-prime. In the general case, a linear congruence as above has a solution if and only if $\gcd(a, n) \mid b$. We now study the analogous problem but for quadratic congruences. We shall first introduce some notation:

3.1. Preliminaries.

Definition 1 (Quadratic Residues). *Let p be a fixed prime. We say an integer x not-divisible by p is a **quadratic residue** modulo p if and only if there exists an integer a such that*

$$x \equiv a^2 \pmod{p} \tag{11}$$

Definition 2 (Legendre⁴ Symbol). *Given a prime integer p may define an operation*

$$\left(\frac{\cdot}{p}\right) : \mathbb{N} \rightarrow \{0, \pm 1\}, \quad n \mapsto \begin{cases} 0 & \text{if } \gcd(n, p) \neq 1 \\ 1 & \text{if } n \text{ is a quadratic residue mod } p \\ -1 & \text{else} \end{cases}$$

⁴Adrien-Marie Legendre was a French mathematician who made numerous contributions to polynomial theory and elliptic equations. His papers on polynomials would later inspire Galois.

The symbol $\left(\frac{\cdot}{p}\right)$ is hereby referred to by the Legendre symbol⁵.

Lemma 11. Let $p \geq 3$ be a prime and define a mapping $\Gamma : (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \{\pm 1\}$ by sending

$$x \mapsto \left(\frac{x}{p}\right)$$

The map Γ is a **surjective** homomorphism of groups under multiplication.

PROOF. We know that $(\mathbb{Z}/p\mathbb{Z})^\times$ is a group under multiplication. This is well defined, since $\gcd(a, x) = 1$ for all $a \in (\mathbb{Z}/p\mathbb{Z})^\times$. Recall from Theorem 8 that this group in question is cyclic, and thus has a generator g such that $(\mathbb{Z}/p\mathbb{Z})^\times = \langle g \rangle$. That is,

$$(\mathbb{Z}/p\mathbb{Z})^\times = \left\{ g, g^2, g^3, \dots, g^{\frac{p-1}{2}}, g^{\frac{p+1}{2}}, \dots, g^{p-1} \right\} \quad (12)$$

We characterize the “squares” of this group by squaring all the elements above:

$$\{g^2, g^4, \dots, g^{p-1}\}$$

Which are precisely the powers of g of even power (p is odd). Similarly, the odd powers of g are therefore not-squares and map to -1 in the target under Γ . Certainly, this implies that Γ is a surjection as was asserted. It remains to show that it is a group homomorphism. The identity element of the group, 1 , is always a square and hence $\Gamma(1) = 1$. The multiplicative property of Γ follows from the fact that the sum of any two odd numbers is even, the sum of evens is even and the sum of an even and an odd is odd.

This concludes the proof of this lemma. ○

The following theorem is due to Gauss:

Theorem 12. Let $p, q \in \mathbb{Z}$ be primes with $p \neq q$. Then,

$$\left(\frac{p}{q}\right) = (-1)^{\frac{(p-1)(q-1)}{2}} \left(\frac{q}{p}\right) \quad (13)$$

Furthermore,

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} \text{ and } \left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8} \\ -1 & \text{if } p \equiv \pm 3 \pmod{8} \end{cases} \quad (14)$$

The proof of the above theorem is postponed, we have yet to prove the required lemmata. In contrast to the previous, this next result is attributed to Leonhard Euler.

Proposition 13 (Eulers Criterion). Let p be an odd prime and a an integer such that $p \nmid a$. Then,

$$\left(\frac{a}{p}\right) = 1 \text{ if and only if } a^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

PROOF. First observe that $\phi : (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times$ given by $\phi(a) := a^{\frac{p-1}{2}}$ is necessarily a group homomorphism by properties of exponentiation over $\mathbb{Z}$. Recall the mapping Γ from Lemma 11, which we know to be a homomorphism of groups. We claim, first, that $\ker \Gamma \subseteq \ker \phi$. Indeed, if $\Gamma(a) = 1$ then, a is a square modulo p . Hence,

$$\phi(a) = a^{\frac{p-1}{2}} = (b^2)^{\frac{p-1}{2}} = b^{p-1} \stackrel{\text{Fermat}}{\equiv} 1 \pmod{p}$$

⁵We denote by “.” an arbitrary natural number.

Let us recall from the proof of Lemma 11 that $2\#\ker\Gamma = \#(\mathbb{Z}/p\mathbb{Z})^\times = p-1$. Now we use the fact that the kernels of homomorphisms are groups in their own right. From the inclusion above (namely $\ker\Gamma \subseteq \ker\phi$) and Lagrange's theorem for groups it follows that either $\ker\phi = \ker\Gamma$ or that $\phi \equiv 1$ on $(\mathbb{Z}/p\mathbb{Z})^\times$. This last option is not possible since then the polynomial

$$f(x) := x^{\frac{p-1}{2}} - 1$$

would have $p-1$ roots, over $(\mathbb{Z}/p\mathbb{Z})^\times$ which is absurd. Hence, $\ker\phi = \ker\Gamma$ which completes the proof by definition of Γ . ○

There exists a highly applicable corollary to the proposition above:

Corollary 14. *Given an odd prime p the congruence relation $x^2 \equiv a \pmod{p}$ has no solution if and only if $a^{\frac{p-1}{2}} \equiv -1 \pmod{p}$.*

PROOF. This follows from the fact that by Fermat's little theorem $a^{\frac{p-1}{2}}$ is its own inverse modulo p and cannot be 1 by Proposition 13. ○

In particular, it follows from Proposition 13 and Corollary 14 that

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \text{ reduced modulo } p \quad (\mathcal{Q})$$

3.2. Gaussian Summations. Gaussian summations are an algebraic concept that will be essential to our proof of Theorem 12. We must first make some recollections. We say a number $\zeta \in \mathbb{C}$ is an n^{th} root of unity provided $\zeta^n = 1$. This root of unity is said to be primitive if this n is the smallest possible n for this to hold true. We introduce the Gaussian summation now:

Definition 3. *Consider an odd prime p . Given an integer a the **Gauss sum at a** is described by*

$$g_a := \sum_{\eta=1}^{p-1} \left(\frac{\eta}{p}\right) \zeta^{a\eta}, \quad \zeta = e^{2\pi i/p} \quad (15)$$

Lemma 15. *For all $a \in \mathbb{Z}$:*

$$\sum_{n=0}^{p-1} \zeta^{an} = \begin{cases} p & \text{if } a \equiv 0 \pmod{p} \\ 0 & \text{else} \end{cases} \quad (16)$$

PROOF. If $a \equiv 0$ modulo p then $a = kp$ for an integer k . In this case⁶, $\zeta^a = (e^{\frac{2\pi i}{p}})^{kp} = e^{2k\pi i} = 1$. Since there are p terms in the summation above, we deduce that $\sum_{n=0}^{p-1} \zeta^{an} = p$. Else, assume that $a \not\equiv 0$ modulo p . Since $a = kp + \alpha$ for $\alpha < p$, we have by properties of the complex exponential that $\zeta^a = \zeta^{\frac{2\pi i \alpha}{p}} \neq 1$ recognizing this as a geometric sum one finds that $\sum_{n=0}^{p-1} \zeta^{an} = \frac{1-\zeta^{ap}}{1-\zeta^a} = \frac{1-1}{\dots} = 0$. ○

⁶This follows from the identity $e^{i\vartheta} = \cos \vartheta + i \sin \vartheta$ for $\vartheta \in \mathbb{R}$ together with (15)

Note that letting $a = x - y$ in the above we obtain the following

$$\sum_{n=0}^{p-1} \zeta^{n(x-y)} = \begin{cases} p & \text{if } x \equiv y \pmod{p} \\ 0 & \text{else} \end{cases}$$

Now we have that $g_0 = \sum_{n=0}^{p-1} \left(\frac{n}{p}\right) = 0$. Certainly, the mapping $\left(\frac{\cdot}{p}\right) : (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \{\pm 1\}$ is known by Lemma 11 to be a surjective homomorphism of groups. Moreover, we recall from the proof that exactly half the elements map to $\{1\}$ and similarly for $\{-1\}$. Consequently, we see that the summation in question vanishes,

Theorem 16. *Let $p \nmid a$. Then,*

$$g_a^2 = (-1)^{\frac{p-1}{2}} p \tag{17}$$